It uses public-key authentication and digital signatures. Until 1970s, there were only symmetric cryptosystems in which transmitter and receiver must have the same key. This raises the problem of key exchange and key management. Unlike a symmetric encryption, the communicating parties need not know other's private-key in asymmetric encryption. Each user generates their own key pair, which consists of a private key and a public key. A public-key encryption method is a method of converting a plaintext with a public key into a ciphertext from which the plaintext can be retrieved with a private key.

17.4.2 Authentication Technology

The main role of security certification is to ensure Authentication, Integrity and Non-repudiation. This can be achieved through digital signatures and digital certificates.

Symmetric Key Encryption	Asymmetric Key Encryption
Same key is used for both encryption and decryption	Different keys are used for encryption and decryption
Speed of encryption or decryption is very fast	Speed of encryption or decryption is comparatively slow
Plain text and cipher text are of same size	The size of cipher text is always greater than plain text.
Algorithms like DES, AES, RC4 uses symmetric key encryption	Algorithms like RSA, ECC, DSA use asymmetric key encryption
Provides confidentiality	Provides confidentiality, authenticity and non-repudiation
The number of key used grows exponentially with the number of users	The number of key used grows linearly with the number of users
	

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